Boundary value problems

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Initial value problem

- Consider a second order differential equation $F(x, y, y', y'') = 0$.
- Its general solution contains two arbitrary constants.
- To determine these constants, we need to prescribe two conditions, which are called **initial conditions** when y and y' are specified at a particular value of x.
- The differential equation together with the initial conditions is called an **initial value problem (IVP)**.

Example: Initial Value Problem (IVP)

Problem

Solve the initial value problem:

$$y'' + y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

Solution

- Auxiliary equation: $m^2 + 1 = 0 \;\Rightarrow\; m = \pm i$
- General solution: $y(x) = c_1 \cos x + c_2 \sin x$
- From $y(0) = 1$: $c_1 = 1$
- From $y'(0) = 0$: $c_2 = 0$

$$\therefore\; y(x) = \cos x$$

Boundary value problem

- If y or y' or their contribution is prescribed at two different values of x, then the condition is called boundary conditions and the differential equation along with **boundary conditions** is called a **boundary value problem**.

Note: The solution of an IVP in general exists and is unique, but the solution of a BVP may exist or may not.

Example

1. Consider

$$y'' + y = 0,\ \ y(0) = 0,\ \ y(\pi) = 1$$

Example

2. Consider

$$y'' + y = 0,\ \ y(0) = 0,\ \ y(\pi) = 0$$

Example

3. Consider

$$y'' + y = 0, \; y(0) = 0, \; y\left(\frac{\pi}{2}\right) = 1$$

Solution of BVP-Finite difference method

By Taylor series explansion of $y(x)$, we have

$$y_{i+1} = y(x_i + h) = y_i + hy_i' + \frac{h^2}{2!}y_i'' + \frac{h^3}{3!}y_i''' + \cdots \tag{1}$$

$$y_{i-1} = y(x_i - h) = y_i - hy_i' + \frac{h^2}{2!}y_i'' - \frac{h^3}{3!}y_i''' + - \cdots \tag{2}$$

By subtraction equation (2) from (1), we get

$$y_i' = \frac{y_{i+1} - y_{i-1}}{2h} \tag{3}$$

By adding equation (1) and (2), we get

$$y_i'' = \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \tag{4}$$

- Now consider the finite difference method of solving linear BVP of the form

$$p(x)y'' + q(x)y' + r(x)y = f(x) \tag{5}$$

- We consider two types of boundary conditions
  1. The boundary conditions do not involve y'
  2. The boundary conditions involve y'

Consider (by setting $x = x_i$ in (5))

$$p_i {y_i}'' + q_i {y_i}' + r_i y_i = f_i. \tag{6}$$

Here, $p_i = p(x_i)$, $q_i = q(x_i)$, $r_i = r(x_i)$, $f_i = f(x_i)$, $y_i = y(x_i)$, ${y_i}' = y'(x_i)$ and ${y_i}'' = y''(x_i)$

Let $y(a) = y_a$ and $y(b) = y_b$

Replace y_i' and ${y_i}''$ in (6) by (3) and (4). This implies,

$$p_i \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + q_i \left[\frac{y_{i+1} - y_{i-1}}{2h} \right] + r_i y_i = f_i, \tag{7}$$

$i = 1, 2, \ldots, n-1$

The equation (7) gives rise to system of equations of order $n - 1$.

Example

1. Solve $xy'' + y = 0$, $y(1) = 1$, $y(2) = 2$ with $h = 0.5$ and $h = 0.25$.

Example

2. Solve $(x^3+1)y'' + x^2y' - 4xy = 2$, $y(0)=0$, $y(2)=4$ with $h=0.5$.

Solution: Here, $h=0.5$. Therefore, $x_0=0$, $x_1=0.5$, $x_2=1$, $x_3=1.5$ and $x_4=2$. Also $y_0=y(0)=0$ and $y_4=y(2)=4$. We will get three equations,

When $i=1$,

$$-11y_1 + 4.75y_2 = 2$$

When $i=2$,

$$7y_1 - 20y_2 + 9y_3 = 2$$

When $i=3$,

$$15.25y_2 - 41y_3 = -77$$

After solving above three equations, we get $y_1 = y(x_1) = y(0.5) = 0.25$, $y_2 = y(x_2) = y(1) = 1$ and $y_3 = y(x_3) = y(1.5) = 2.25$

The boundary conditions involve y'

Let the boundary conditions be

$$\alpha_o y(a) + \alpha_1 y'(a) = \alpha_2$$

$$\beta_o y(b) + \beta_1 y'(b) = \beta_2$$

Changing the derivatives into finite differences, we get

$$\alpha_o y_o + \alpha_1 y'_o = \alpha_2$$

$$\beta_o y_n + \beta_1 y'_n = \beta_2$$

Consider,

$$y_i' = \frac{y_{i+1} - y_{i-1}}{2h}.$$

This implies

$$y_o' = \frac{y_1 - y_{-1}}{2h}$$

and

$$y_n' = \frac{y_{n+1} - y_{n-1}}{2h}.$$

Therefore,

$$\alpha_o y_o + \alpha_1 \left[\frac{y_1 - y_{-1}}{2h}\right] = \alpha_2 \tag{8}$$

$$\beta_o y_n + \beta_1 \left[\frac{y_{n+1} - y_{n-1}}{2h}\right] = \beta_2 \tag{9}$$

Here the problem involves finding y_{-1} and y_{n+1} (the values of y outside the interval of interest)
Then equation (8) will enable us to find y_{-1} in terms of y_o and y_1 and equation (9) will enable us to find y_{n+1} in terms of y_{n-1} and y_n.

Example

1. Solve $xy'' + y = 0$, $y'(1) = 0$, $y(2) = 1$ with $h = 0.5$

Example

2. Solve $y'' + (1+x)y' - y = 0$, $y(0) = y'(0)$, $y(1) + y'(1) = 1$ with $h = 0.5$.

Consider

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + F(x, y, u, u_x, u_y) = 0. \tag{10}$$

Equation (10) is

1. elliptic if $B^2 - AC < 0$
2. parabolic if $B^2 - AC = 0$
3. hyperbolic if $B^2 - AC > 0$

Elliptic equations

1. Laplace equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$
2. Poisson's equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = F(x, y)$

Solution to Laplace's and Poisson's equation in a rectangular region

1. Divide the region into small squares by drawing lines parallel to the sides of the rectangle
2. The points of the intersection of these grid lines are called mesh points or lattice points or grid points.
3. To solve Laplace's equation, $u_{ij} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1})$
   To remember easily $u_{ij} = \frac{1}{4}(N + E + W + S)$
4. To solve Poisson's equation,
   $u_{ij} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - h^2 f_{ij})$
   To remember easily $u_{ij} = \frac{1}{4}(N + E + W + S - h^2 f_{ij})$

Parabolic equations

1. Heat equation: $\frac{\partial u}{\partial t} = c\frac{\partial^2 u}{\partial x^2}$
   $u(x, 0) = f(x)$
   $u(a, t) = \phi(t),\ u(b, t) = \psi(t),\ t > 0$
2. Divide $[a, b]$ into n sub-intervals each of length h. That is, $h = \frac{b-a}{n}$.
3. The values of grid points at bottom edge of the rectangle can be obtained using $u_{io} = u(x, 0) = f(x_i)$.
4. The remaining values can be obtained by $u_{i,j+1} = \frac{u_{i+1,j}+u_{i-1,j}}{2}$
5. Time step is obtained by $k = \frac{h^2}{3c}$

Hyperbolic equation

- Wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, a < x < b, t > 0 \tag{11}$$

$$u(x,0) = f(x), \frac{\partial u}{\partial t}(x,0) = g(x) \tag{12}$$

$$u(a,t) = \phi(x), u(b,t) = \psi(x) \tag{13}$$

- Divide $[a,b]$ into n sub-intervals each of length h. That is, $h = \frac{b-a}{n}$.
- The values of grid points at bottom edge of the rectangle can be obtained using $u_{io} = u(x,0) = f(x_i)$.
- The values of u at first time step is given by $u_{i1} = \frac{1}{2}(u_{i+1,0} + u_{i-1,0}) + kg_i$, where $g_i = \frac{\partial u}{\partial t}(x,0)$
- The remaining values can be obtained by $u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}$

- The Laplace's equation says, at every interior point, the value of u equals the average of its neighboring values when there are no sources/sinks inside the domain.
- The Poisson's equation says, at every interior point, the value of u is influenced by the average of its neighbors plus the effect of sources/sinks inside the domain.
- The heat equation describes how heat energy flows in a medium over time.
- The wave equation describes how disturbances propagate through a medium.